Question - 8: Write a Python program to implement Depth First Search (DFS).

Program Code

```
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)


graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("DFS Traversal:")
dfs(graph, 'A', set())
```

Program Output

```
PS E:\Desktop\SIMATS\A
ython/Python314/python
y"
DFS Traversal:
A B D E C F
PS E:\Desktop\SIMATS\A
```